# SBAN v4: Elastic Regional Synaptic Birth-Death Assembly Networks in Zig

A non-transformer online byte-learning research artifact with runtime growth and shrink targets, region-parallel sparse lanes, and conservative bridge-memory scaffolding on enwik8

**Core outcome.** On the packaged 30k x 4 enwik8 protocol, the best SBAN v4 prefix model is **sban_v4_4bit** at **40.51%**, beating the order-2 baseline by **1.04 pp**. The best drift model is **sban_v4_6bit** at **42.94%**, beating order-2 by **2.95 pp**. The largest reliable scientific gain remains local self-rating: removing reputation hurts the 4-bit model by **3.08 pp** on prefix and **3.20 pp** on drift. On a separate hard-to-easy elasticity probe, the default 4-bit model grows its short-memory target from **2048** to **8192** and then shrinks to **4608**, while its live short memories collapse from more than **4718** to only **198**.

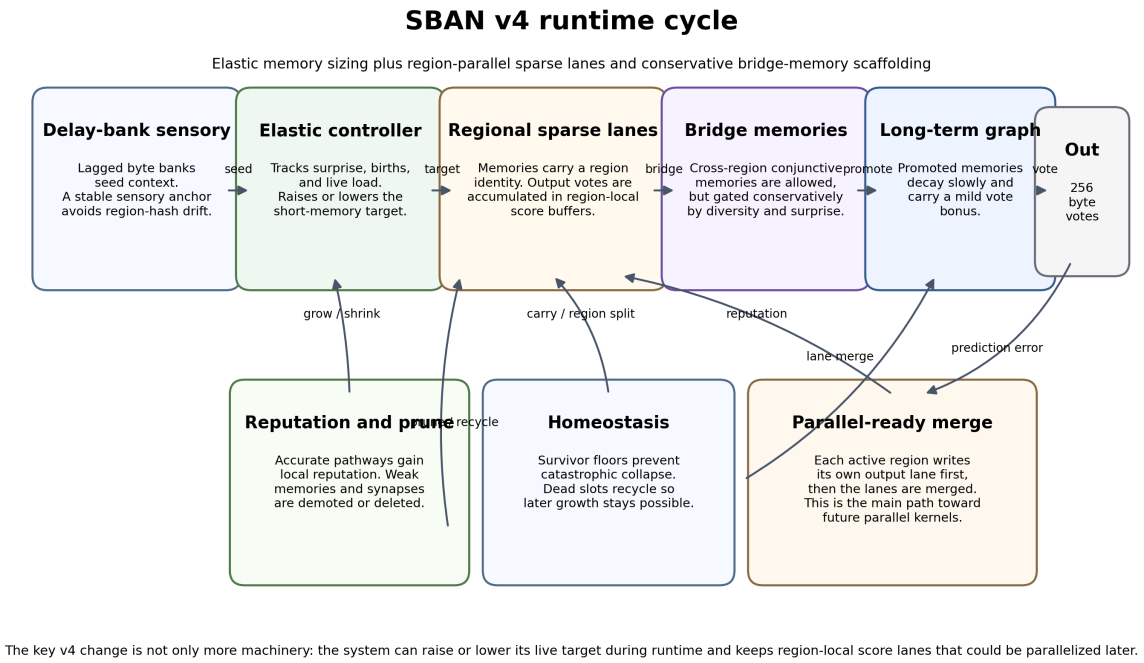


***Figure.*** *SBAN v4 adds an elastic short-memory target, region-local score lanes, and conservative bridge-memory scaffolding on top of the earlier online sparse graph.*

## 1. Motivation and v4 design changes

SBAN v3 already established a dual-memory, reputation-gated sparse graph that could learn enwik8 online, but the next research question was architectural scaling rather than only another bit sweep. The user asked for a system that can expand when the stream demands more internal structure and contract when the stream becomes simpler, while also moving toward a more parallel execution path and a more explicit notion of graph substructure. SBAN v4 therefore keeps the v3 local-learning core but rewrites large parts of the runtime around three new ideas: an elastic short-memory target, region-tagged sparse lanes, and a conservative bridge-memory mechanism for cross-region conjunctions.

A practical design issue appeared during early v4 experiments: sensory tokens should not be reassigned to different regions every time the region count changes. Doing that destabilizes regional identity. The

implemented v4 scaffold therefore uses a stable sensory anchor and lets new regions emerge from overloaded memory subgraphs rather than from a moving token hash. This makes the regional split a load-balancing and consolidation mechanism, not a changing partition of the raw byte alphabet.

The v4 bridge mechanism is intentionally conservative. It allows a memory to declare both a primary and a secondary region, but only under diversity plus surprise conditions. That choice was made after testing a more eager bridge rule, which created too many cross-region nodes and hurt accuracy. The present v4 artifact should therefore be read as a stronger engineering scaffold for future bridge research rather than as a final solved bridge design.

## 2. Experimental protocol

The main evaluation remains fully prequential: at each byte the model predicts the next byte and then updates immediately with the observed target. The packaged rebuild runs four experiment groups. First, a 30k x 4 prefix sweep evaluates 120k contiguous byte predictions on enwik8. Second, a 30k x 4 drift sweep concatenates windows from offsets 0, 25M, 50M, and 75M with transient state reset at boundaries while preserving learned structure. Third, a 40k x 4 longer prefix stress run evaluates the 4-bit ablations. Fourth, a synthetic elasticity probe concatenates 60k real enwik8 bytes with a 60k low-entropy repeated tail so the controller experiences both a hard regime and an easy regime within one run.

The baselines remain online order-1 and order-2 byte predictors. As in earlier SBAN artifacts, the model emits vote scores rather than calibrated probabilities, so the primary metric is top-1 online accuracy; top-5 is reported to separate candidate coverage from final winner selection. The repo also keeps v3 reference JSON so the new scaffold can be compared against the immediately previous artifact under the same 30k x 4 protocol.

| Model | Prefix top-1 | Drift top-1 | Prefix top-5 | Drift top-5 |
|---|---|---|---|---|
| sban_v4_1bit | 39.70% | 42.06% | 69.89% | 69.95% |
| sban_v4_2bit | 40.28% | 42.54% | 72.29% | 72.39% |
| sban_v4_3bit | 40.43% | 42.84% | 72.77% | 73.09% |
| sban_v4_4bit | 40.51% | 42.86% | 72.57% | 72.91% |
| sban_v4_5bit | 40.49% | 42.91% | 72.64% | 72.78% |
| sban_v4_6bit | 40.33% | 42.94% | 72.34% | 72.68% |
| sban_v4_7bit | 40.09% | 42.78% | 72.16% | 72.44% |
| sban_v4_8bit | 40.27% | 42.67% | 72.18% | 72.34% |
| markov_order2 | 39.46% | 39.99% | 80.96% | 79.86% |

*Table 1. SBAN v4 bit sweep on the packaged 30k x 4 prefix and drift protocols. Order-1 is omitted from the discussion below because it stays far below all SBAN variants.*

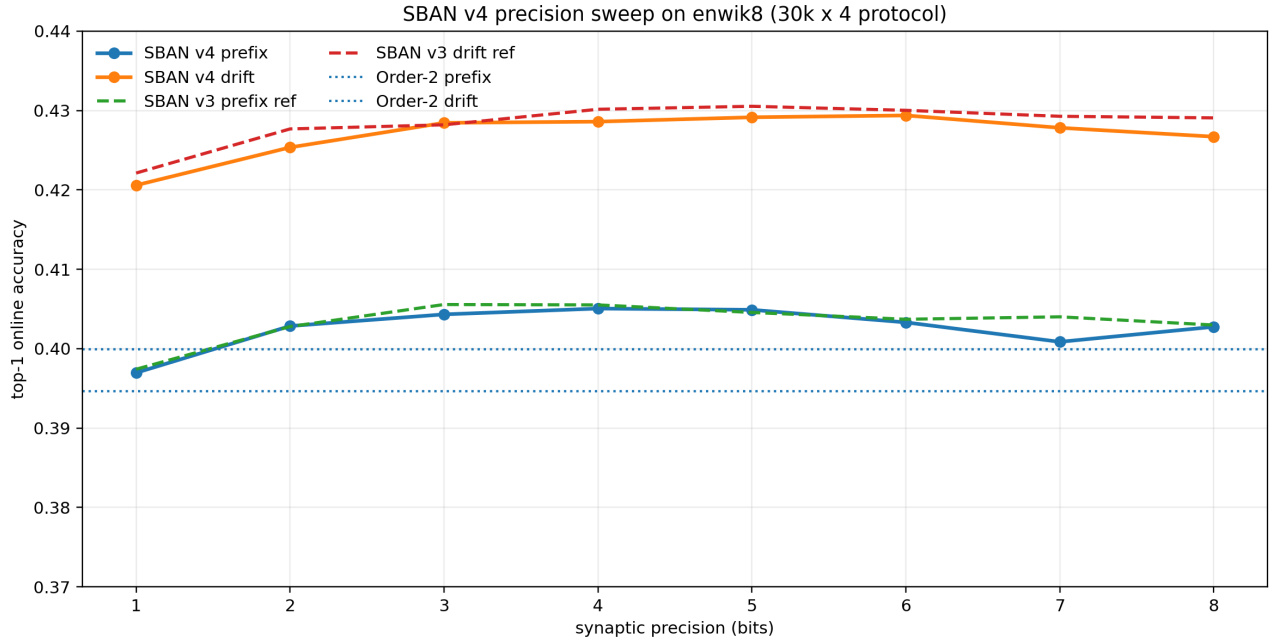


***Figure.*** *Precision scaling for SBAN v4 compared with the bundled v3 reference on the same 30k x 4 protocol. The broad plateau remains, with drift strongest near 5-6 bits.*

# 3. What v4 adds scientifically, and what it costs

| Comparison | Prefix top-1 | Drift top-1 |
|---|---|---|
| SBAN v3 best reference | 40.56% | 43.05% |
| SBAN v4 best | 40.51% | 42.94% |
| Delta (v4 - v3) | -0.05 pp | -0.11 pp |
| SBAN v3 4-bit | 40.55% | 43.02% |
| SBAN v4 4-bit | 40.51% | 42.86% |
| 4-bit delta (v4 - v3) | -0.05 pp | -0.16 pp |

*Table 2. Direct v3-to-v4 comparison on the same 30k x 4 protocol. v4 adds runtime elasticity and regional scaffolding, but the current default does not yet beat v3 on raw top-1.*

The comparison is mixed in a scientifically useful way. v4 keeps the earlier sparse non-transformer behavior and adds a real growth-and-shrink control path, but its best top-1 results are slightly below the bundled v3 reference. That means the new machinery is not a free accuracy gain. The right interpretation is that v4 is an architectural scaffold for dynamic scaling, not yet the new best static operating point. This matters because the user asked for a system that changes size during runtime; that requirement forces a different engineering tradeoff than simply tuning the older v3 graph for maximum top-1.

The good news is that the main v4 mechanisms are at least competitive. The default 4-bit elastic model is slightly ahead of the fixed-capacity 4-bit ablation on both prefix and drift, and almost tied with the single-region ablation. The bridge mechanism is the least mature part: on prefix the no-bridge ablation is slightly better, while on drift the default regains a very small lead. In other words, the present bridge rule is safe enough not to collapse the system, but still too weak to become a major source of gain on enwik8.

| Model | Prefix top-1 | Drift top-1 | Regions | Target short | Bridge births |
|---|---|---|---|---|---|
| sban_v4_4bit | 40.51% | 42.86% | 7 | 8192 | 1 |
| sban_v4_4bit_no_bridge | 40.60% | 42.81% | 7 | 8192 | 0 |
| sban_v4_4bit_fixed_capacity | 40.34% | 42.78% | 1 | 4096 | 0 |
| sban_v4_4bit_single_region | 40.52% | 42.85% | 1 | 8192 | 0 |
| sban_v4_4bit_no_reputation | 37.42% | 39.66% | 7 | 8192 | 0 |
| markov_order2 | 39.46% | 39.99% | 0 | 0 | 0 |

*Table 3. The main v4 4-bit ablations. Reputation remains the largest determinant of decisive top-1 accuracy. Bridge births are intentionally sparse in the current scaffold.*

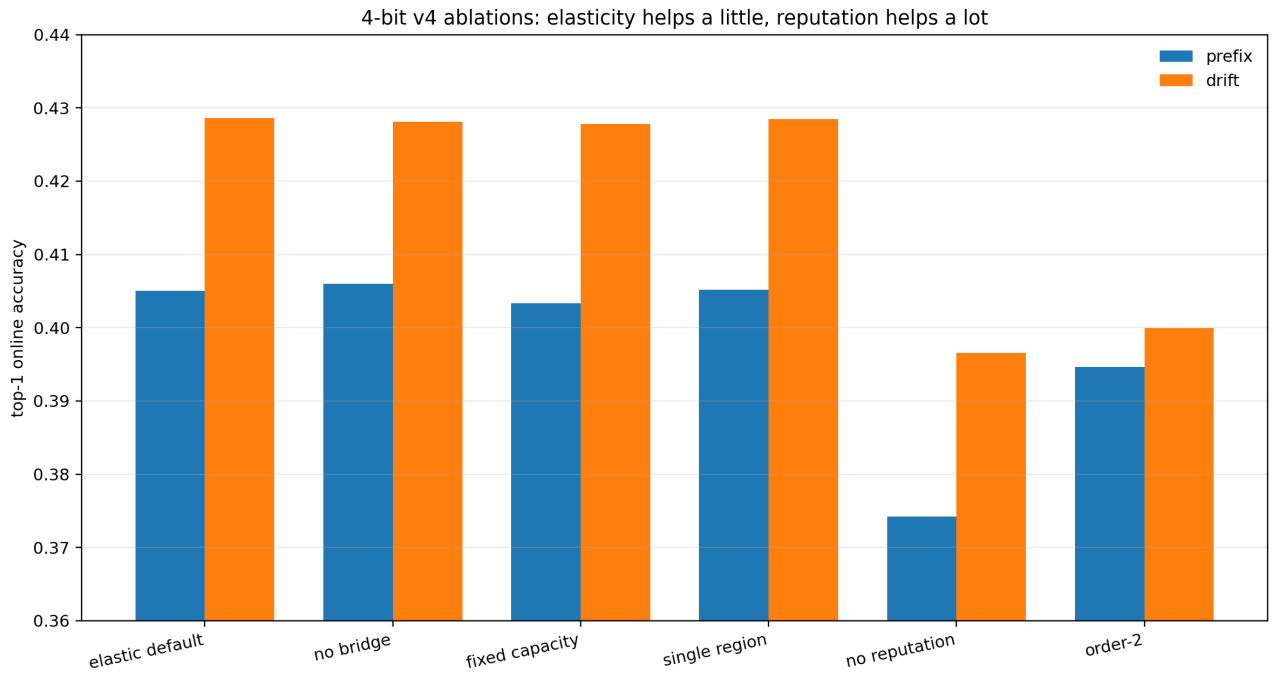


***Figure.*** *Removing reputation produces the largest collapse. The elastic regional default is slightly ahead of fixed capacity, while the bridge rule is close to neutral on the main enwik8 protocol.*

# 4. Longer contiguous prefix stress

| Model | Top-1 | Top-5 | Final memories | Regions | Target short | Synapses |
|---|---|---|---|---|---|---|
| sban_v4_4bit | 42.48% | 73.99% | 1963 | 7 | 8192 | 28424 |
| sban_v4_4bit_no_bridge | 42.42% | 73.99% | 1913 | 7 | 8192 | 27740 |
| sban_v4_4bit_fixed_capacity | 42.54% | 74.07% | 2291 | 1 | 4096 | 31260 |
| sban_v4_4bit_single_region | 42.46% | 73.94% | 1972 | 1 | 8192 | 29164 |
| sban_v4_4bit_no_reputation | 41.53% | 91.63% | 2120 | 7 | 8192 | 22377 |
| markov_order2 | 41.32% | 81.66% | 0 | 0 | 0 | 0 |

*Table 4. Longer 40k x 4 contiguous prefix stress. The elastic default stays above order-2 by more than one point, but the fixed-capacity variant is slightly ahead on this specific longer stress run.*

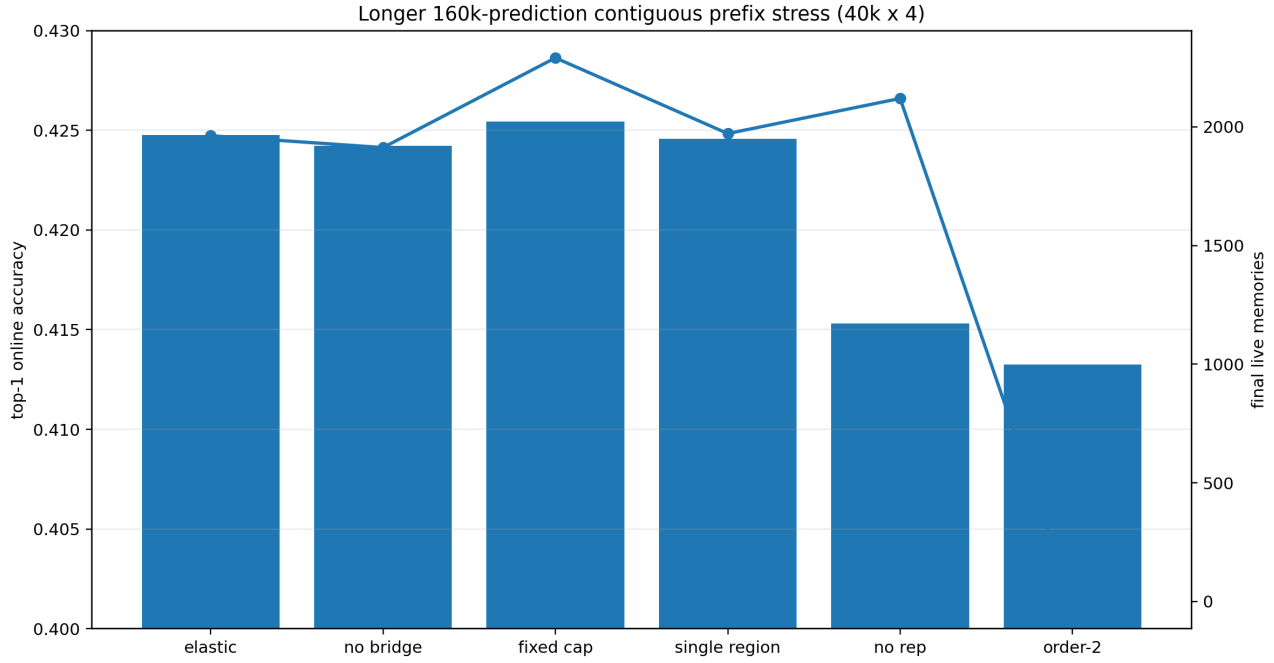


***Figure.*** *Longer 160k-prediction prefix stress. The default elastic model remains above the order-2 baseline while ending with fewer than two thousand live memories, but fixed capacity is still a very competitive ablation.*

The longer prefix stress is useful because it separates two questions that are easy to blur together: does the architecture stay adaptive, and does the elasticity controller automatically become the best choice? The answer here is mixed. The default 4-bit elastic model still beats order-2 by **1.15 pp** and ends with only **1963** live memories, yet the fixed-capacity ablation edges it out by about **0.07 pp**. So v4 elasticity is clearly viable, but not yet the universally best controller on every long run.

# 5. Elasticity probe: direct evidence of runtime growth and shrink

| Model | Accuracy | Grows | Shrinks | Final target | Final short | Regions |
|---|---|---|---|---|---|---|
| sban_v4_4bit | 69.89% | 12 | 28 | 4608 | 198 | 7 |
| sban_v4_4bit_no_bridge | 70.02% | 12 | 28 | 4608 | 198 | 7 |
| sban_v4_4bit_fixed_capacity | 69.22% | 0 | 0 | 4096 | 192 | 1 |
| sban_v4_4bit_single_region | 69.35% | 12 | 28 | 4608 | 191 | 1 |
| sban_v4_4bit_no_reputation | 67.75% | 12 | 28 | 4608 | 198 | 7 |
| markov_order2 | 69.51% | 0 | 0 | 0 | 0 | 0 |

*Table 5. Elasticity probe. The default model expands aggressively during the hard half of the stream and then contracts when the stream becomes simple, while fixed capacity obviously cannot change target size at all.*

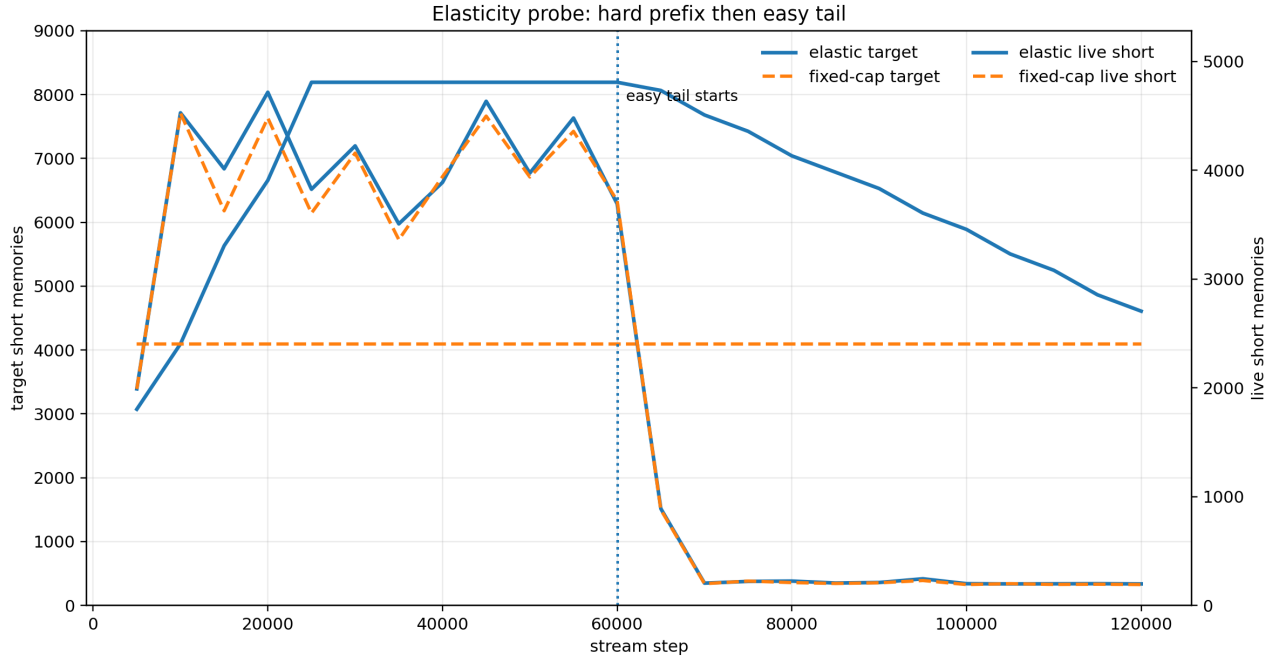


***Figure.*** *A hard-to-easy probe shows the main v4 controller behavior directly. The elastic default grows to 8192 target short memories on the difficult half, then shrinks to 4608 on the easy tail while live short memories collapse from more than 4600 to roughly 200.*

This probe is the strongest direct evidence that v4 actually satisfies the user request for runtime size adaptation. The default 4-bit model executes **12** growth events and **28** shrink events, reaches a peak target of **8192**, and ends at **4608**. It also slightly beats the fixed-capacity 4-bit ablation by **0.67 pp** and the order-2 baseline by **0.37 pp**. The important caveat is that region allocation does not yet compact: live memories shrink hard, but the region scaffold itself remains allocated across seven live regions.

# 6. Interpretation and remaining limitations

SBAN v4 is therefore best understood as a scientific machine-learning scaffold rather than a final solved architecture. It demonstrates four things clearly: online local self-improvement still works under the new controller, the graph can expand and contract its target size during runtime, regional score lanes are compatible with the sparse graph, and synaptic reputation remains essential for suppressing bad-habit consolidation. Those are meaningful steps toward the user goal of a dynamic, brain-inspired non-transformer model.

At the same time, the new bridge subsystem is not yet paying for itself on the main enwik8 sweep. Bridge births are extremely sparse under the current conservative gate, and the no-bridge ablation is slightly stronger on prefix. The region scaffold also grows but does not compact, so the controller currently changes target size and live memory count more than it changes the structural footprint of the region layout. Finally, v4 does not yet surpass v3 on the main benchmark, so the present result is an engineering broadening rather than a pure accuracy win.

The highest-value next steps are therefore specific. First, bridge memories need a better selection rule, probably one that depends on region-specific error attribution rather than only diversity and surprise. Second, regions need compaction or merge rules so the scaffold can truly shrink structurally rather than only through memory death. Third, the elasticity controller itself is still hand-tuned; a learned or meta-optimized controller may outperform the current threshold schedule. Fourth, the low-bit synapses are still not bit-packed in RAM, and the scores are still not calibrated probabilities, so hardware efficiency and compression metrics remain future work.

# 7. Reproducibility

The packaged script rebuilds the Zig binary, reruns the main prefix and drift sweeps, reruns the 4-bit ablations and long stress run, generates the elasticity probe, redraws the figures, rewrites the executive summary and README, and regenerates this PDF. The commands below assume the current repository root and a compatible Zig compiler.

```
zig build test
zig build -Doptimize=ReleaseFast
./zig-out/bin/zig_sban eval-enwik data/enwik8 docs/results/enwik_prefix_v4.json prefix 30000 5000 4096
./zig-out/bin/zig_sban eval-enwik data/enwik8 docs/results/enwik_drift_v4.json drift 30000 5000 4096
./zig-out/bin/zig_sban eval-ablations data/enwik8 docs/results/enwik_prefix_ablation_v4.json prefix 4 30000 5000
4096
./zig-out/bin/zig_sban eval-ablations data/enwik8 docs/results/enwik_drift_ablation_v4.json drift 4 30000 5000
4096
./zig-out/bin/zig_sban eval-ablations data/enwik8 docs/results/enwik_prefix_long_ablation_v4.json prefix 4 40000
5000 4096
python scripts/make_artifacts.py --zig /path/to/zig --dataset /path/to/enwik8.zip
```

**References**


[1] Mahoney, M. Large text compression benchmark corpus (enwik8).

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[4] Benna, M. K., and Fusi, S. (2016). Computational principles of synaptic memory consolidation. Nature Neuroscience.

[5] Courbariaux, M., Bengio, Y., and David, J.-P. (2015). BinaryConnect.